MILITARY SPECIFICATION

VALVE, STEAM PRESSURE REDUCING,

QUIET DESIGN

This amendment forms a part of Military Specification MIL-V-24517(SHIPS), 28 April 1975 and is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

3.3, Table I, under "Materials", opposite "Cylinder, liner and piston": Delete "440°C" and substitute "440C".

PAGE 3

3.4.7, third sentence: Delete.

PAGE 4

3.4.10.1.1, delete and substitute:

"3.4.10.1.1 Quiet-flow regulating element. Element design employed to achieve a low-noise pressure loss shall avoid excessively small flow passages or any other feature jeopardizing operational reliability. The design and method of fabrication and inspection of the quiet-flow regulating element shall assure that the cross-sectional area at the point of greatest restriction of any given flow passage shall not vary by more than plus or minus 3 percent on a unit-per-unit basis from the nominal point and that the overall flow capacity of the regulating element shall not vary by more than plus or minus 5 percent on a unit-per-unit basis. It shall be brazed or otherwise fabricated into a one-piece configuration capable of withstanding installation and operational stresses. Internal passages shall be such that under any production tolerance condition 0.033-inch diameter perforated metal screen type strainer installed upstream will protect against clogging. Element cleaning shall not be required more often than annually. Replacement due to blockage, erosion, wear, etc. not more often than once every 5 years. The element shall be capable of being cleaned onboard ship, either by backflush or with ultrasonic equipment. Any special fixtures necessary for cleaning of the element shall be supplied with each valve. The element structure shall be of a solid or continuous nature, not susceptible to media migration."

3.4.10.1.2, line 7: Delete "lever" and substitute "liner".

3.4.10.2, third sentence: Delete.

3.5, item (b): Delete "250 to 740" and substitute: "315 to 850".

PAGE 5

3.6.4, line 2: Delete "130 to 200" and substitute "150 to 180".

3.6.5, add: "Structureborne noise levels at frequencies above the 3.15 kHz 1/3 octave band of up to 10 dB above this requirement are acceptable for flow rates above 7,500 lb/hr."

PAGE 6

3.8.1, add as item (h):

"(h) Calculations verifying compliance with the flow passage and overall flow-rate consistency requirements specified in 3.4.10.1.1. Required dimensional inspection shall be detailed."

PSC 4820

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:19Z
Check the source to verify that this is the current version before use.

PAGE 7

4.3.1, third sentence after listing: Delete and substitute: "This valve shall also have an accelerometer block welded to the body and an accelerometer block welded to the outlet flange."

4.3.5, line 2: Delete "130" and substitute "150".

4.3.7.1, line 1: Delete "250" and substitute "315".

4.3.7.1, line 2: Delete "130" and substitute "150".

PAGE 8

4.3.7.2, line 2: Delete "200" and substitute "180".

4.3.7.4, lines 1 and 4: Delete "250" and substitute "315".

Add as paragraph 4.4.1.1:

"4.4.1.1 The quiet-flow regulating element shall have undergone, during production, the inspection detailed in 3.8.1(h). A complete record of the inspection results for each element shall be maintained by the manufacturer and made available for Government inspection for at least 5 years after delivery of the valve (or element)."

PAGE 9

4.4.6: Delete:

PAGE 12

Figure 1: Delete.

NOTE: The margins of this amendment are marked "#" to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH
(Project 4820-N370)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:19Z
Check the source to verify that this is the current version before use.